Input: $X \subseteq [n]^M$

Output: A partition $X = X^1 \cup X^2 \cup \cdots \cup X^m$

Initialize $i := 1$.

while $X \neq \emptyset$ **do**

  Let $I \subseteq J$ be a maximal subset (possibly $I = \emptyset$) such that $\mathrm{H}_\infty(\boldsymbol{X}(I)) < \gamma|I| \log n$ and let $\alpha_i \in [n]^I$ be a witness of this fact, i.e., $\mathbf{Pr}[\boldsymbol{X}(I) = \alpha_i] > n^{-\gamma|I|}$.

  $X^i := \{x \in X : x(I) = \alpha_i\}$ and $I_i := I$.

  Update $X := X \setminus X^i$, $J := J \setminus I_i$, and $i := i + 1$.